# 6.4 Exponential Functions

Name _____ Period ____ Date _____

Algebra I 6.4 · Student sheet

**I can…**

- identify and evaluate exponential functions.
- graph exponential functions and find their asymptote, domain, and range.
- tell whether a table or situation is linear or exponential.

**Vocabulary** · fill in the missing word

**Exponential function** — $y =$ _____, where $a \neq 0$, $b \neq 1$, and $b > 0$

**Asymptote** — a line a graph gets closer and closer to but never _____

**Linear vs. exponential** — linear: add the same _____ each step; exponential: multiply by the same amount

## Part 1 · Savings plans

Plan A: save \$10 on day 1, \$20 on day 2, \$30 on day 3, and so on. Plan B: save \$1 on day 1, \$2 on day 2, \$4 on day 3, doubling each day.

| Day | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Plan A (\$) | 10 | 20 | 30 | | | | | | | |
| Plan B (\$) | 1 | 2 | 4 | | | | | | | |

Plan A: each day I _____. Plan B: each day I _____.

Which plan saves more on day 10? Which would you choose for 4 days?

## Part 2 · Notes and examples (fill in each note, then work the example)

**Exponential function:** An **exponential function** has the form $y =$ _____, where $a \neq 0$, $b > 0$, and $b \neq 1$.

### Example 1

Does each equation represent an exponential function?

a. $y = 4(7)^x$ b. $y = -6x$ c. $y = x^3$

**Linear or exponential tables:** Linear tables have a constant _____. Exponential tables have a constant _____.

### Example 2

Is each table linear or exponential?

a. x: 0, 1, 2, 3 y: 3, 6, 12, 24

b. x: 0, 1, 2, 3 y: 1, 4, 7, 10

### Example 3

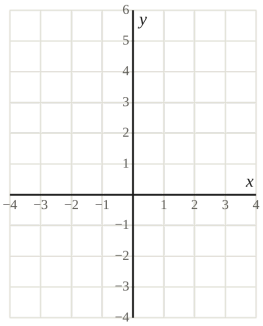
Evaluate each function.

a. $y = -2(5)^x$ when $x = 3$ b. $y = 3(\frac{1}{2})^x$ when $x = -2$

**Graphs:** The graph of $y = ab^x + k$ has an asymptote at _____. The domain is _____.

### Example 4

Graph $y = 2^x$ and $y = 2^x - 3$. Find the y-intercept, asymptote, domain, and range of $y = 2^x - 3$.




**Exponential situations:** A situation is exponential when it changes by the same _____ each time.

### Example 5

Is each situation linear or exponential?

a. A tank loses 7 gallons of water every hour.

b. A town's population grows 3% every year.

## Part 3 · Partner practice

### Partner 1

Evaluate $y = 4(3)^x$ when $x = 2$.

**Partner 2**

For $y = 3^x + 2$, find the y-intercept, asymptote, and range.